



$$(d) F_4 = x'y' + xz$$

2.4.1 Algebraic Manipulation

- A literal is a primed or unprimed variable.
- The minimization of the number of literals and the number of terms results in a circuit with less equipment.
- Minimization criterion for now: Literal minimization
- The number of literals in a Boolean function can be minimized by algebraic manipulations.

Note: There are no specific rules to follow that will guarantee the final answer.

Example 2-2: Simplify the following Boolean functions to a minimum number of literals

$$\begin{aligned}
 1. x + x'y &= (x + x'y) + x'y \quad (\text{By Theorem 4(a)}) \\
 &= x + (xy + x'y) \quad (\text{By Theorem 4(a)}) \\
 &= x + (y(x + x')) \quad (\text{Postulate 3(b)}) \\
 &= x + y \cdot (x + x') \quad (\text{Postulate 4(a)}) \\
 &= x + y \cdot 1 \quad (\text{Postulate 5(a)}) \\
 &= x + y \quad (\text{Postulate 2(b)})
 \end{aligned}$$

$$\begin{aligned}
 2. x(x' + y) &= x \cdot x' + x \cdot y \quad (\text{Postulate 4(a)}) \\
 &= 0 + xy \quad (\text{Postulate 5(b)}) \\
 &= xy + 0 \quad (\text{Postulate 3(a)}) \\
 &= xy \quad (\text{Postulate 2(a)})
 \end{aligned}$$

$$\begin{aligned}
 3. x'y'z + x'yz + xy' &= z \cdot (x'y') + z \cdot (x'y) + xy' \quad (\text{Postulate 3(b)}) \\
 &= (z \cdot x') \cdot y' + (z \cdot x') \cdot y + xy' \quad (\text{Theorem 4(b)}) \\
 &= (z \cdot x') \cdot (y' + y) + xy' \quad (\text{Postulate 4(a)}) \\
 &= (z \cdot x') \cdot 1 + xy' \quad (\text{Postulate 5(a)}) \\
 &= (zx') + xy' \quad (\text{Postulate 5(a)})
 \end{aligned}$$